

>> myScript

Iter	Eval	Objective	Objective	BestSoFar	BestSoFar	BoxConstraint	KernelScale	Standardize
	result	runtime	(observed)	(estim.)				

1	Best	0.125	1.6883	0.125	0.125	612.42	257.86	true
2	Accept	0.46429	0.33046	0.125	0.14921	0.0045532	66.605	false
3	Accept	0.14286	0.11331	0.125	0.12556	0.17702	20.419	true
4	Accept	0.46429	0.10406	0.125	0.12541	3.4442	0.0068929	false
5	Best	0.125	0.18955	0.125	0.12472	996.4	15.544	true
6	Accept	0.46429	0.050558	0.125	0.12503	994.67	1.7209	true
7	Accept	0.17857	0.036414	0.125	0.12504	0.0019641	123.93	true
8	Accept	0.17857	0.044766	0.125	0.12505	0.10516	654.39	true
9	Accept	0.125	0.031268	0.125	0.12504	24.518	10.791	true
10	Accept	0.46429	0.042982	0.125	0.12504	0.0024535	0.0010018	true
11	Accept	0.46429	0.044167	0.125	0.12504	225.63	0.67945	false
12	Accept	0.17857	0.052473	0.125	0.12496	0.011808	338.43	true
13	Accept	0.125	0.047579	0.125	0.12472	13.315	13.093	true
14	Accept	0.125	0.052007	0.125	0.12495	23.471	52.539	true
15	Accept	0.125	0.049759	0.125	0.12503	56.881	90.609	true
16	Accept	0.30357	0.060077	0.125	0.12501	147.19	995.97	false
17	Accept	0.5	0.068766	0.125	0.12501	0.0023264	0.02248	true
18	Accept	0.14286	0.057323	0.125	0.1251	0.0011642	9.5014	true
19	Accept	0.14286	0.065029	0.125	0.1254	979.51	58.381	true
20	Accept	0.14286	0.065326	0.125	0.12603	999.02	142.21	true

Iter	Eval	Objective	Objective	BestSoFar	BestSoFar	BoxConstraint	KernelScale	Standardize
	result	runtime	(observed)	(estim.)				

21	Accept	0.125	0.069153	0.125	0.12536	25.235	14.642	true
22	Accept	0.46429	0.050397	0.125	0.12527	223.96	0.001043	false
23	Accept	0.5	0.046533	0.125	0.12516	221.19	6.6582	false
24	Accept	0.46429	0.037255	0.125	0.12503	218.99	0.068992	false
25	Accept	0.5	0.032788	0.125	0.12489	0.0018708	0.18414	true
26	Accept	0.125	0.039045	0.125	0.12459	22.493	15.325	true
27	Accept	0.17857	0.031054	0.125	0.12475	21.05	250.46	true
28	Accept	0.14286	0.028252	0.125	0.12718	19.809	19.115	true
29	Accept	0.46429	0.030878	0.125	0.12725	18.955	0.0044457	true
30	Accept	0.44643	0.030767	0.125	0.12719	214.72	260.9	false

Optimization completed.

MaxObjectiveEvaluations of 30 reached.

Total function evaluations: 30
 Total elapsed time: 12.1841 seconds
 Total objective function evaluation time: 3.5903

Best observed feasible point:

<u>BoxConstraint</u>	<u>KernelScale</u>	<u>Standardize</u>
996.4	15.544	true

Observed objective function value = 0.125
 Estimated objective function value = 0.12652
 Function evaluation time = 0.18955

Best estimated feasible point (according to models):

<u>BoxConstraint</u>	<u>KernelScale</u>	<u>Standardize</u>
25.235	14.642	true

Estimated objective function value = 0.12719
 Estimated function evaluation time = 0.050513

Training samples : 56
 Testing samples : 23
 The KNN accuracy by 3 k is 69.57%
 The SVM Accuracy: 86.96%
 >> myScript

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Iter	Eval	Objective	Objective	BestSoFar	BestSoFar	BoxConstraint
KernelScale		Standardize				
	result		runtime	(observed)	(estim.)	
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1	Best	0.5	0.15552	0.5	0.5	0.010821
0.0026932	false					
2	Best	0.17857	0.048331	0.17857	0.2298	0.0030739
5.6863	true					
3	Accept	0.5	0.035616	0.17857	0.19842	143.5
0.015601	true					
4	Accept	0.35714	0.036008	0.17857	0.19541	4.1024
145.44	false					
5	Accept	0.33929	0.031614	0.17857	0.28304	0.0031069
2.4972	true					
6	Accept	0.19643	0.031487	0.17857	0.17866	0.0030568
6.3468	true					
7	Accept	0.23214	0.031195	0.17857	0.17869	988.89
5.074	true					
8	Accept	0.19643	0.033019	0.17857	0.18638	0.0010048
4.8171	true					
9	Accept	0.17857	0.038511	0.17857	0.17829	0.0010314
5.5914	true					

10	Accept	0.46429	0.030672	0.17857	0.1783	0.0010233
0.61156	true					
11	Accept	0.19643	0.031188	0.17857	0.18497	0.0065141
5.4984	true					
12	Accept	0.21429	0.037787	0.17857	0.18637	0.0010251
12.838	true					
13	Accept	0.17857	0.035588	0.17857	0.18357	0.0010096
7.31	true					
14	Accept	0.19643	0.029789	0.17857	0.18667	0.0010054
7.0909	true					
15	Accept	0.21429	0.032808	0.17857	0.18674	0.0010933
44.846	true					
16	Accept	0.21429	0.030134	0.17857	0.18662	0.0026628
126.32	true					
17	Best	0.16071	0.033	0.16071	0.16193	893.98
460.76	true					
18	Accept	0.21429	0.034147	0.16071	0.16231	0.0012988
948.66	true					
19	Accept	0.21429	0.040534	0.16071	0.16393	0.0010005
376.3	true					
20	Accept	0.19643	0.031054	0.16071	0.17183	972.83
627.21	true					

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| Iter | Eval | Objective | Objective | BestSoFar | BestSoFar | BoxConstraint|
KernelScale | Standardize |
|      | result |          | runtime   | (observed) | (estim.)   |
|      |         |          |           |             |             |
|      |         |          |           |             |             |
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21	Best	0.125	0.030864	0.125	0.12834	988.55
273.5	true					
22	Best	0.071429	0.037094	0.071429	0.083069	998.16
199.36	true					
23	Accept	0.125	0.031401	0.071429	0.088956	996.16
107.51	true					
24	Best	0.035714	0.033966	0.035714	0.055409	986.81
174.69	true					
25	Accept	0.035714	0.028464	0.035714	0.042051	990
164.76	true					
26	Accept	0.035714	0.031004	0.035714	0.038523	956.74
158.43	true					
27	Accept	0.035714	0.029045	0.035714	0.03745	996.5
160.35	true					
28	Accept	0.035714	0.030298	0.035714	0.036217	969.77
155.57	true					
29	Accept	0.23214	0.03694	0.035714	0.036165	635.56
995.77	false					
30	Accept	0.5	0.035566	0.035714	0.036345	992.46
0.11026	true					

Optimization completed.

Best observed feasible point:

BoxConstraint	KernelScale	Standardize
986.81	174.69	true

Observed objective function value = 0.035714
Estimated objective function value = 0.041495
Function evaluation time = 0.033966

Best estimated feasible point (according to models):

BoxConstraint	KernelScale	Standardize
996.5	160.35	true

Estimated objective function value = 0.036345
Estimated function evaluation time = 0.031931

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Training samples : 56
Testing samples : 23
The KNN accuracy by 9 k is 69.57%
The SVM Accuracy: 86.96%
>> myScript
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Iter	Eval	Objective	Objective	BestSoFar	BestSoFar	BoxConstraint
KernelScale	Standardize					
	result		runtime	(observed)	(estim.)	
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1	Best	0.5	0.062743	0.5	0.5	0.010821
0.0026932	false					
2	Best	0.17857	0.038261	0.17857	0.2298	0.0030739
5.6863	true					
3	Accept	0.5	0.03924	0.17857	0.19842	143.5
0.015601	true					
4	Accept	0.35714	0.033612	0.17857	0.19541	4.1024
145.44	false					
5	Accept	0.17857	0.035736	0.17857	0.17856	0.0032382
5.7861	true					
6	Accept	0.21429	0.032093	0.17857	0.17857	0.1893
18.046	true					
7	Accept	0.42857	0.03498	0.17857	0.17845	30.012
2.0334	true					
8	Accept	0.21429	0.030597	0.17857	0.17861	0.012499
135.67	true					

	9		Accept		0.5		0.032919		0.17857		0.17842		0.0052738	
0.0026804			true											
	10		Accept		0.21429		0.027946		0.17857		0.17829		0.0013301	
20.976			true											
	11		Accept		0.19643		0.0339		0.17857		0.17841		0.014641	
9.5704			true											
	12		Accept		0.46429		0.028379		0.17857		0.17815		0.0010081	
1.6829			true											
	13		Accept		0.21429		0.032096		0.17857		0.17815		0.0043541	
61.966			true											
	14		Accept		0.21429		0.031776		0.17857		0.17815		0.0012083	
532.5			true											
	15		Accept		0.21429		0.034077		0.17857		0.17816		0.001857	
999.4			true											
	16		Accept		0.5		0.032653		0.17857		0.17816		3.9564	
5.4879			false											
	17		Accept		0.41071		0.027467		0.17857		0.17816		3.4648	
997.06			false											
	18		Accept		0.5		0.028745		0.17857		0.17816		2.4667	
0.1181			false											
	19		Accept		0.19643		0.027874		0.17857		0.17844		36.536	
6.6033			true											
	20		Accept		0.19643		0.030114		0.17857		0.18608		0.001015	
6.5267			true											

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	Iter		Eval		Objective		Objective		BestSoFar		BestSoFar		BoxConstraint	
KernelScale			Standardize											
			result				runtime		(observed)		(estim.)			

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	21		Accept		0.19643		0.042901		0.17857		0.18922		0.0010056	
6.6881			true											
	22		Best		0.16071		0.031056		0.16071		0.16075		179.58	
265.21			true											
	23		Accept		0.21429		0.029137		0.16071		0.18871		11.461	
223.61			true											
	24		Accept		0.21429		0.029908		0.16071		0.189		9.6711	
315.84			true											
	25		Accept		0.16071		0.03094		0.16071		0.18053		0.14463	
8.2269			true											
	26		Accept		0.17857		0.028512		0.16071		0.18014		3.7075	
8.7004			true											
	27		Accept		0.5		0.027745		0.16071		0.1801		0.37651	
0.13769			true											
	28		Accept		0.46429		0.027633		0.16071		0.18007		0.005999	
0.79987			false											
	29		Accept		0.5		0.027863		0.16071		0.18001		204.93	
0.017283			false											
	30		Accept		0.46429		0.028636		0.16071		0.17997		0.0089926	
29.497			false											

Optimization completed.
MaxObjectiveEvaluations of 30 reached.
Total function evaluations: 30
Total elapsed time: 4.1562 seconds
Total objective function evaluation time: 0.97954

Best observed feasible point:

BoxConstraint	KernelScale	Standardize
179.58	265.21	true

Observed objective function value = 0.16071
Estimated objective function value = 0.19848
Function evaluation time = 0.031056

Best estimated feasible point (according to models):

BoxConstraint	KernelScale	Standardize
0.14463	8.2269	true

Estimated objective function value = 0.17997
Estimated function evaluation time = 0.032136

Training samples : 56
Testing samples : 23
The KNN accuracy by 7 k is 65.22%
The SVM Accuracy: 91.30%
>> myScript

Iter	Eval	Objective	Objective	BestSoFar	BestSoFar	BoxConstraint	KernelScale	Standardize
	result	runtime	(observed)	(estim.)				
1	Best	0.5	0.16242	0.5	0.5	0.010821	0.0026932	false
2	Best	0.17857	0.034847	0.17857	0.2298	0.0030739	5.6863	true
3	Accept	0.5	0.029118	0.17857	0.19842	143.5	0.015601	true
4	Accept	0.35714	0.033741	0.17857	0.19541	4.1024	145.44	false
5	Accept	0.33929	0.031667	0.17857	0.28304	0.0031069	2.4972	true
6	Accept	0.19643	0.034473	0.17857	0.17867	0.0030549	6.2043	true
7	Accept	0.23214	0.033816	0.17857	0.18905	975.48	4.9898	true
8	Accept	0.19643	0.035985	0.17857	0.19089	0.0010418	5.1767	true
9	Accept	0.19643	0.028841	0.17857	0.18701	0.014428	9.3212	true
10	Accept	0.21429	0.034113	0.17857	0.18827	0.38179	27.015	true
11	Accept	0.21429	0.031702	0.17857	0.18834	0.0010536	82.669	true
12	Accept	0.17857	0.033489	0.17857	0.18892	805.39	283.68	true
13	Accept	0.21429	0.03222	0.17857	0.19002	0.001276	661.4	true
14	Accept	0.21429	0.033217	0.17857	0.18975	0.0010063	246.95	true
15	Best	0.16071	0.033152	0.16071	0.16641	983.92	404.34	true

16	Accept	0.16071	0.034013	0.16071	0.16307	988.75	441.93	true
17	Accept	0.16071	0.030603	0.16071	0.16167	992.78	450.04	true
18	Accept	0.46429	0.030888	0.16071	0.16169	0.0017329	999.39	false
19	Accept	0.16071	0.032709	0.16071	0.16122	999.57	454.73	true
20	Accept	0.21429	0.031233	0.16071	0.1613	951.81	939.23	true
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Iter	Eval	Objective	Objective	BestSoFar	BestSoFar	BoxConstraint	KernelScale	Standardize
	result		runtime	(observed)	(estim.)			
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21	Accept	0.5	0.033999	0.16071	0.16129	961.21	12.221	false
22	Accept	0.46429	0.034115	0.16071	0.1613	0.0042947	0.0010053	true
23	Accept	0.5	0.028078	0.16071	0.16136	0.001029	0.17194	true
24	Accept	0.5	0.030639	0.16071	0.1614	0.0010226	1.5378	false
25	Accept	0.16071	0.030274	0.16071	0.16113	992.7	446.91	true
26	Accept	0.5	0.028537	0.16071	0.16116	987.52	0.25842	false
27	Accept	0.5	0.028756	0.16071	0.1612	998.36	0.029961	false
28	Accept	0.46429	0.027753	0.16071	0.16121	0.0010403	41.494	false
29	Accept	0.46429	0.028067	0.16071	0.16126	172	0.003827	true
30	Accept	0.5	0.027748	0.16071	0.1613	947.47	0.001007	false

Optimization completed.
MaxObjectiveEvaluations of 30 reached.
Total function evaluations: 30
Total elapsed time: 5.2797 seconds
Total objective function evaluation time: 1.0802

Best observed feasible point:

BoxConstraint	KernelScale	Standardize
983.92	404.34	true

Observed objective function value = 0.16071
Estimated objective function value = 0.16197
Function evaluation time = 0.033152

Best estimated feasible point (according to models):

BoxConstraint	KernelScale	Standardize
988.75	441.93	true

Estimated objective function value = 0.1613
Estimated function evaluation time = 0.032228

Training samples : 56

Testing samples : 23

The KNN accuracy by 7 k is 65.22%

The SVM Accuracy: 86.96%

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